When Does Bayes by Backprop Know What It Doesn't Know?

A transient out-of-distribution window, and how to find it

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October 2026

Abstract

Bayesian neural networks are often said to detect out-of-distribution (OOD) inputs better than deterministic networks, but their predictions are also softer everywhere, and a deterministic network can be softened for free by a temperature. I compare each uncertainty method with a deterministic network softened to the same in-distribution entropy, in a series of pre-registered experiments. On MLPs trained on MNIST, Bayes by Backprop (BBB) at typical training budgets is no better than this softness-matched network. Its OOD advantage appears only in a *transient window* of training. Starting from small posterior variances, the KL term slowly widens the posterior. For a while BBB beats the control by +0.06 to +0.08 AUROC, more than a deep ensemble. Then the ELBO keeps improving towards a near-prior posterior that underfits and loses the advantage. Whether a reported BBB model sits in this window depends on the number of optimizer steps, not on the amount of data, and neither held-out NLL nor the ELBO finds it. Selecting the checkpoint that best separates validation images from synthetic outliers recovers 73% of the best achievable advantage without any real OOD data. On CIFAR-10 with a small CNN no window appears: there, neither BBB nor a deep ensemble beats the softness-matched network on SVHN. The same MLP on CIFAR-10 gains at most +0.05 on CIFAR-100 for a few thousand steps before it collapses towards the prior. On natural images, the window is small or absent for both architectures.

1 Introduction

A standard classifier gives a confident prediction for any input, including inputs unlike its training data. Bayesian neural networks (BNNs) average predictions over a distribution of weights [1], which should raise uncertainty on unfamiliar inputs. The usual evidence is that a BNN's predictive entropy is higher on OOD data than a deterministic network's.

That evidence has a confound. A BNN's averaged prediction is softer than a deterministic network's on every input, and temperature scaling [2] softens a deterministic network at no cost. Temperature is a known ingredient of OOD detection [3] and appears as a baseline in uncertainty benchmarks [4]. Whether BNNs are intrinsically good OOD detectors has been questioned [5], and a Bayesian last layer is known to reduce overconfidence far from the data [6]. The comparison I use throughout is simple: each method against a deterministic network softened, on validation data only, to that method's own in-distribution entropy. The question is whether the method ranks OOD inputs as more uncertain than in-distribution inputs better than a network of the same softness.

Findings.

1. At standard training budgets, mean-field BBB is no better than the softness-matched network on MNIST → Fashion-MNIST and KMNIST. Deep ensembles are slightly better; MC dropout ties; last-layer Laplace is worse (§3.1).
2. BBB's advantage is set by the number of optimizer steps, not by the training-set size (§3.2).
3. It is a transient: it opens as posterior variances grow and closes as the posterior approaches the prior, while the ELBO keeps improving. A large variance initialisation prevents learning altogether (§3.3).
4. It needs posterior noise in both hidden layers together, and it is not a pixel shortcut (§3.4).
5. Synthetic outliers built from validation images find the window without real OOD data, recovering 73% of the oracle advantage (§3.5).
6. On CIFAR-10 there is no comparable window, with either a CNN or the same MLP; the drift towards the prior does replicate (§4).

Every experiment had its hypothesis, predictions and falsification criteria committed to git before it ran. Appendix B lists all predictions, including the many that failed.

2 Setup

Data. MNIST training subsets of $n \in \{500, 2{,}000, 10{,}000\}$ class-stratified images (one per seed); a fixed validation pool of 5,000 further training images, disjoint from every subset; the MNIST test set; OOD: Fashion-MNIST [7] and KMNIST [8] test sets, normalised with MNIST statistics. CIFAR-10 experiments are described in §4.

Models. All MNIST models use a 784–256–256–10 ReLU MLP. *DNN*: Adam (10^{-3}, cosine), 40 epochs, batch 64. *MAP*: the same network trained to the MAP estimate under BBB's $\mathcal{N}(0, 1)$ prior, with BBB's budget (150 epochs, gradient clipping). *MC dropout* [9]: $p = 0.1$, 50 samples. *Deep ensemble* [10]: 5 DNNs. *Last-layer Laplace* [11]: full generalised Gauss–Newton covariance over the last layer of a DNN, prior precision tuned on validation NLL. *BBB* [1]: mean-field Gaussian posterior, $\mathcal{N}(0, 1)$ prior, per-example negative ELBO

$$\mathcal{L} = \frac{1}{B} \sum_{i \in \text{batch}} -\log p(y_i \mid x_i, w) + \frac{1}{N} \mathrm{KL}\big[q(w) \,\|\, p(w)\big], \qquad w = \mu + \sigma \odot \varepsilon, \tag{1}$$

with the KL weight warmed up linearly, $\sigma = \mathrm{softplus}(\rho)$ and ρ initialised at -5 ($\sigma_0 \approx 0.007$), Adam 10^{-3}, gradient clipping, 50 posterior samples at test time. By default BBB trains for 150 epochs at $n \geq 2{,}000$ (4,800 steps at $n = 2{,}000$) and the same number of steps at $n = 500$.

The softness-matched control. For a method M, I fit a temperature T to a DNN's logits so that its mean predictive entropy on the *validation* set equals M's. No test or OOD data enters the fit. The reported quantity is $\Delta = \mathrm{AUROC}(M) - \mathrm{AUROC}(\mathrm{DNN}_T)$, where AUROC measures the separation of test from OOD images by predictive entropy. $\Delta > 0$ means M ranks OOD inputs better than a network of the same softness. Comparisons are paired by seed, with bootstrap confidence intervals and Wilcoxon tests.

3 Results on MNIST

3.1 At standard budgets, only ensembles beat the control

Table 1 compares each method with its own softness-matched DNN over 10 seeds. At $n = 500$ and 2,000, BBB is significantly *worse* than the control; at $n = 10{,}000$ it is much better, in all 10 seeds. §3.2 shows that this jump is not about data. Deep ensembles beat the control at every n by +0.01 to +0.02; against a stronger, budget-matched MAP network their edge mostly disappears at $n = 2{,}000$. MC dropout roughly ties the control, and last-layer Laplace is clearly below it.

n	method	Δ Fashion [95% CI]	Δ KMNIST [95% CI]
500	BBB	−0.046 [−0.063, −0.032]	−0.035 [−0.041, −0.028]
500	ensemble	+0.014 [0.003, 0.027]	+0.010 [0.008, 0.012]
500	MC dropout	−0.003 [−0.016, 0.009]	−0.019 [−0.021, −0.018]
500	last-layer Laplace	−0.072 [−0.088, −0.055]	−0.112 [−0.122, −0.102]
2,000	BBB	−0.024 [−0.048, −0.001]	−0.010 [−0.014, −0.007]
2,000	ensemble	+0.015 [0.007, 0.022]	+0.015 [0.013, 0.018]
2,000	MC dropout	+0.007 [0.002, 0.012]	−0.010 [−0.012, −0.007]
2,000	last-layer Laplace	−0.040 [−0.047, −0.032]	−0.043 [−0.047, −0.038]
10,000	BBB	+**0.048** [0.038, 0.059]	+**0.069** [0.065, 0.074]
10,000	ensemble	+0.016 [0.008, 0.026]	+0.023 [0.021, 0.025]
10,000	MC dropout	−0.010 [−0.018, −0.001]	−0.007 [−0.010, −0.003]
10,000	last-layer Laplace	−0.014 [−0.023, −0.006]	−0.012 [−0.016, −0.008]

Table 1: Each method vs a DNN softened to its own validation entropy (entropy AUROC difference, 10 seeds). Source: `experiments/E2_mnist_grid`.

On accuracy, BBB beats the budget-matched MAP network by 1.1 points at $n = 500$ and 0.5 points at $n = 2{,}000$ (9/10 and 10/10 seeds), and loses by 0.5 points at $n = 10{,}000$. About half of its apparent lead over the plain DNN at $n = 2{,}000$ is ordinary regularisation.

3.2 The advantage follows optimizer steps, not data

With fixed epochs, a larger n also means more optimizer steps: 23,550 at $n = 10{,}000$ vs 4,800 at $n \leq 2{,}000$. Figure 1 decouples the two. With the same data, 23,550 steps give $\Delta = +0.06$ to +0.08 in 5/5 seeds at every n, including $n = 500$. 4,800 steps give no advantage, including at $n = 10{,}000$. Over this range the training-set size barely matters.

3.3 A transient window

To see what happens in between, I trained BBB for 50,000 steps at $n = 2{,}000$ with a constant learning rate and evaluated every 1,000 steps (Figure 2). With the default initialisation, Δ is negative for the first ∼3,000 steps. It peaks at +0.067 to +0.074 (KMNIST) around 9,000–14,000 steps, when the mean first-layer σ is 0.65–0.82. It is *negative again* at 25,000–30,000 steps and near zero at 50,000. Meanwhile the negative ELBO keeps falling (143 at 4,000–5,000 steps, 6.8 at 24,000–25,000, 3.5 at 50,000). It falls towards a posterior with $\sigma \approx 0.98$ in the hidden layers (0.93 in the last), accuracy 0.89 and ECE 0.19. This drift towards the prior is what Coker et al. [12] predict for the optimum of wide mean-field BNNs. A slightly larger initial σ (0.049) reaches the window sooner (3,000 steps instead of 6,000). Initialising $\sigma \geq 0.31$ prevents learning altogether:

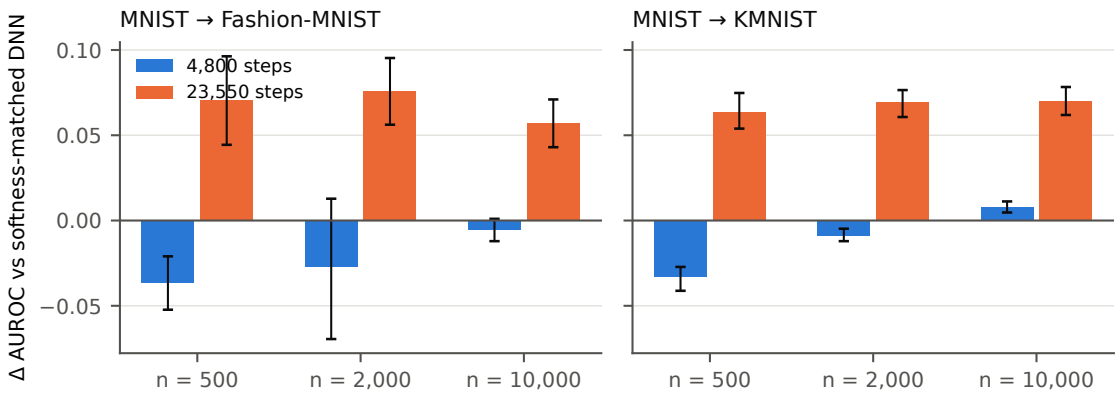


Figure 1: BBB vs its softness-matched DNN with the same training data and two step budgets (5 seeds, 95% CIs). Source: `experiments/E4_steps_vs_n`.

accuracy stays at chance, in a local optimum whose ELBO is far worse (9.0–9.2) than that of the runs that learned.

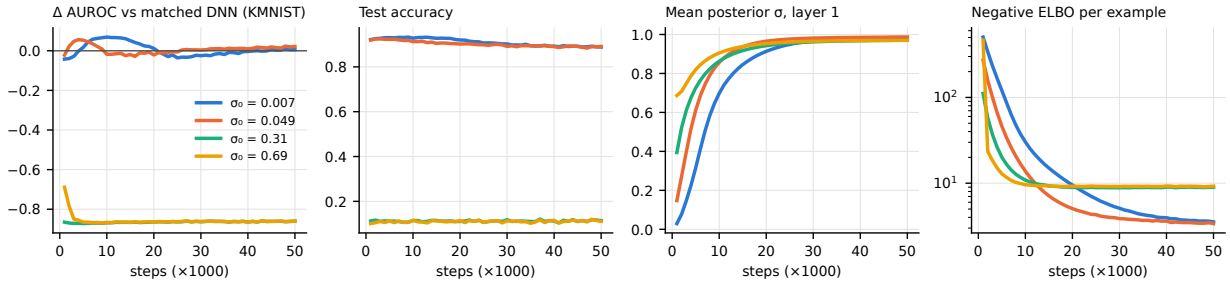


Figure 2: BBB training dynamics at $n = 2{,}000$ (constant learning rate; mean of 3 seeds per initial σ_0). Source: `experiments/E6_dynamics`.

3.4 What produces the window

Noise in both hidden layers. Taking BBB models from the window, I switched posterior noise off layer by layer. Noise in the first and second layers together keeps 73–82% of Δ. Noise in any single layer keeps essentially none (first layer: 16% on Fashion-MNIST, −26% on KMNIST). The posterior-mean network on its own is a worse OOD detector than the DNN. The advantage comes from noise compounding across the hidden layers, which a last-layer method cannot produce.

Not a pixel shortcut. Many pixels are constant (always background) across a training subset: on average 192, 148 and 104 at $n = 500$, 2,000 and 10,000. A model-free rule, "flag any image with ink on such a pixel", separates MNIST from Fashion-MNIST and KMNIST with AUROC 0.967–1.000 over 15 training subsets (means 0.990–1.000), a real weakness of these benchmarks. But the models do not rely on it. Resetting those pixels to background in the OOD images leaves every method's AUROC and Δ essentially unchanged (BBB at $n = 10{,}000$: $+0.057 \to +0.055$ and $+0.070 \to +0.070$). The growth of σ is also global, not concentrated on those pixels (ratio 1.0–1.24).

3.5 Finding the window without OOD data

Neither in-distribution criterion finds the window. Minimum held-out NLL stops at 1,000–3,000 steps, before it; minimum −ELBO stops at the end of training, after it. I built three synthetic outlier sets from the validation images only: uniform noise, digits flipped vertically and horizontally, and digits with one fixed random pixel permutation. I then selected the checkpoint that best separates validation images from these proxies by mutual information (Table 2). Over 12 runs ($n \in \{500, 2{,}000, 10{,}000\}$, two initialisations), the combined proxy rule recovers 73% of the oracle advantage (69–77% in every setting), at the same accuracy and ECE as the oracle checkpoint. No single proxy reaches 70%.

selection rule	mean step	$\bar{\Delta}$	fraction of oracle	accuracy
oracle (uses real OOD)	10,917	+0.067	1.00	0.922
combined proxies	10,417	+0.049	0.73	0.924
flipped digits only	13,750	+0.045	0.67	0.923
minimum −ELBO	39,417	+0.034	0.52	0.906
minimum validation NLL	2,250	−0.032	−0.49	0.922

Table 2: Checkpoint selection ($\bar{\Delta}$ = mean of Fashion-MNIST and KMNIST Δ; 12 runs). Source: `experiments/E9_selection`.

The window opens at a similar step for every n (5,000–8,000). Its peak does not move reliably with n (median 10,000, 11,000 and 11,000 steps). What changes is when it closes: by 14,000–19,000 steps at $n = 2{,}000$ in all six runs, but still open at 40,000 at $n = 10{,}000$. That fits a weaker KL pull ($1/N$) when there is more data. At $n = 500$, two of three windows also stay open, which this explanation does not account for.

4 CIFAR-10

I repeated the dynamics experiment on CIFAR-10 ($n = 10{,}000$, no augmentation) with a small CNN (three 3×3 conv layers of 32, 64 and 128 channels, then 256 and 10 units), every layer Bayesian. OOD sets were SVHN (10,000 images) and CIFAR-100. No window appeared (Figure 3). BBB's peak Δ on SVHN was −0.044, +0.006 and −0.056 in the three seeds, and on CIFAR-100 at most +0.031, early in training. Even a 5-member deep ensemble lost to a single softness-matched DNN on SVHN (AUROC 0.727–0.740 vs 0.759–0.807), while winning on CIFAR-100 by +0.02 to +0.03. The drift towards the prior did replicate. The large fully connected layer's σ reached 0.996, the ELBO kept improving, and accuracy fell from 0.67–0.68 to 0.61–0.63. The first conv layer's σ stayed small (0.04).

To separate architecture from data, I trained the MNIST study's MLP on flattened CIFAR-10 images at $n = 2{,}000$ and 10,000 (3 seeds each). Its deterministic version reaches 38% and 46% test accuracy. BBB again showed no window on SVHN (peak Δ −0.031 to +0.030). On CIFAR-100 it gained +0.030 to +0.046 for the first few thousand steps, similar to the CNN's +0.024 to +0.031. Then it collapsed towards the prior (first-layer $\sigma \to 0.997$–0.999, accuracy falling to 0.10–0.29). Much larger CIFAR-100 "gains" at $n = 2{,}000$ (+0.09 to +0.21) appear only after collapse, at near-chance accuracy, and are not meaningful. My pre-registered decision rule counted these runs as support for an architectural explanation, but it had no accuracy condition, and the qualifying gains are barely above its threshold and no larger than the CNN's. On natural images, neither architecture shows the MNIST-sized window (+0.06 to +0.08 on both OOD sets). The

evidence points more to the data, or to the much harder task (37–47% accuracy), than to the CNN's structure.

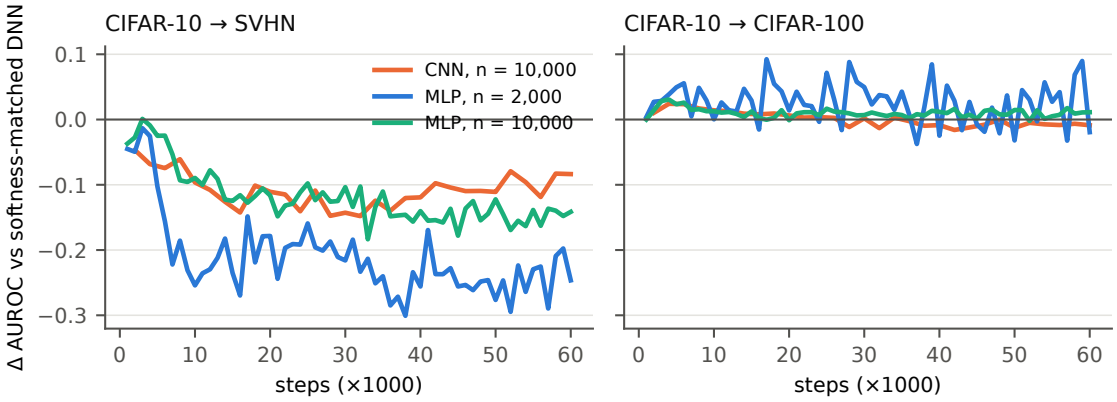


Figure 3: BBB vs its softness-matched network on CIFAR-10 over training (mean of 3 seeds per curve; CNN from all layers Bayesian, MLP = the MNIST architecture on flattened images). Sources: `experiments/E7_cifar`, `experiments/E7b_cifar_mlp`.

5 Related work

Coker et al. [12] prove that the optimal mean-field posterior predictive approaches the prior predictive as width grows; the drift in §3.3 is that effect along the optimisation path of a standard-width network. Unlu and Aitchison [13] note that mean-field variances are often unconverged when training stops; here that state is exactly where BBB looks like a tempered DNN. Izmailov et al. [14] show that weights on always-zero pixels keep their prior under HMC and hurt accuracy under covariate shift; I find no such pixel-specific effect in BBB's OOD detection. Ovadia et al. [4] include temperature-scaled baselines and find ensembles strongest under shift. Henning et al. [5] question whether exact BNN posteriors are intrinsically good OOD detectors. Raina and Schmah [15] report Bayesian advantages for OOD detection from small training sets; the results here suggest checking training length and softness before attributing such advantages to the posterior.

6 Limitations

The window result rests on small MLPs and MNIST-family data. On CIFAR-10, neither a CNN ($n = 10{,}000$, checkpoints every 2,000 steps) nor the same MLP ($n \in \{2{,}000, 10{,}000\}$, every 1,000 steps) showed it, with one initialisation each and no augmentation. I cannot tell whether the window depends on the data or on how hard the task is for these small models. The default hyperparameters (learning rate, $\mathcal{N}(0, 1)$ prior, batch size) were not tuned per method. The prior scale in particular should move the window, and I have not swept it. I did not compare with an exact posterior (e.g. HMC), so I cannot say whether the window resembles exact Bayesian behaviour or is purely a mean-field artefact. The selection rule was tested only on MNIST, where it recovers 73% of the oracle; it failed on CIFAR-10, where there was no window to find.

7 Conclusion

For Bayes by Backprop, "knowing what it doesn't know" is not a property of the trained model. It is a property of where training stopped. At typical budgets, BBB is a tempered network; trained far longer, it drifts towards the prior; in between, on MLPs with MNIST-family data, it detects OOD inputs better than a softness-matched network or a deep ensemble. Comparisons that give BBB the same epochs as a deterministic network, or that stop it on held-out NLL, will not see this. Synthetic outliers are a cheap way to find the window when it exists.

Reproducibility. Code: `bnnstudy/`; every command: `COMMANDS.md`; every number in this report: `CLAIMS.md`; pre-registered protocols with results: `experiments/`.

A KL scaling

Dividing the KL term in Eq. 1 by the number of mini-batches instead of by N is an easy slip when the cross-entropy is averaged per batch. It over-weights the KL term by $N/\lceil N/B \rceil \approx B$ (62.5× at $N = 2{,}000$, $B = 64$). In a 3-seed pilot it cost BBB 3.2 points of accuracy (0.904 vs 0.936) and raised ECE from 0.022 to 0.185.

B Pre-registered predictions and outcomes

Each protocol (`experiments/<id>/protocol.md`) was committed before its runs. Of the 29 entries below, 9 were confirmed, 16 falsified, 3 had mixed outcomes and 1 had no outcome. (E5's P1 held by construction and is omitted.) The falsified ones are what moved the study from "is the advantage just softness?" to the training-window account.

exp.	prediction	outcome
E1	P0: new pipeline reproduces old numbers (±0.02 acc, ±0.03 AUROC)	acc/ECE yes; AUROC off by 0.035
E1	P1: correct KL weight raises BBB accuracy	confirmed
E1	P2: BBB within 0.02 AUROC of softness-matched DNN	falsified (Fashion)
E2	P1: BBB never beats the control	falsified ($n = 10{,}000$)
E2	P2: ensembles beat the control	confirmed
E2	P3: MC dropout and Laplace within 0.02 of the control	dropout yes; Laplace falsified
E2	P4: larger KL weight lowers accuracy, raises KMNIST MI-AUROC	confirmed
E3	P1: BBB and budget-matched MAP equally accurate (< 0.005)	falsified (narrowly)
E3	P2: BBB never beats softness-matched MAP	falsified ($n = 10{,}000$)
E4	P1: the gain follows optimizer steps, not n	confirmed
E4	P2: dead-pixel weights have $> 5\times$ the σ of live ones	falsified
E4	P3: $n = 500$ with long training shows the gain	confirmed
E5	P2: removing dead-pixel ink halves BBB's gain	falsified
E5	P3: removing it shrinks the ensemble's gain	falsified
E5	P4: the DNN's AUROC drops less than BBB's when the ink is removed	falsified (neither drops)
E6	P1: Δ rises with σ	falsified (peaks, then falls)
E6	P2: Δ plateaus	falsified (transient)
E6	P3: larger σ_0 reaches the gain sooner	falsified ($\sigma_0 \geq 0.31$ never learns)
E6	P4: ELBO still improving at 4,800 steps	confirmed
E6b	P1: first-layer noise alone keeps $\geq 70\%$	falsified
E6b	P2: last-layer noise alone keeps $< 30\%$	confirmed
E9	P1: combined proxies reach $\geq 70\%$ of oracle; NLL $\leq 20\%$	confirmed
E9	P2: one proxy alone reaches $\geq 70\%$	falsified (67%)
E9	P3: peak step rises with n	confirmed by mean; not robust
E7	P1: window on SVHN with a CNN	falsified
E7	P2: window closes	no outcome (no window)
E7	P3: proxy selection beats NLL	falsified
E7	P4: ensemble beats the control on SVHN	falsified
E7b	rule: architecture vs data	"architecture" by the rule; weak

References

[1] Charles Blundell, Julien Cornebise, Koray Kavukcuoglu, and Daan Wierstra. Weight uncertainty in neural networks. In *International Conference on Machine Learning (ICML)*, 2015.

[2] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning (ICML)*, 2017.

[3] Shiyu Liang, Yixuan Li, and R. Srikant. Enhancing the reliability of out-of-distribution image detection in neural networks. In *International Conference on Learning Representations (ICLR)*, 2018.

[4] Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua V. Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model's uncertainty?

Evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[5] Christian Henning, Francesco D'Angelo, and Benjamin F. Grewe. Are Bayesian neural networks intrinsically good at out-of-distribution detection? In *ICML Workshop on Uncertainty and Robustness in Deep Learning (UDL)*, 2021.

[6] Agustinus Kristiadi, Matthias Hein, and Philipp Hennig. Being Bayesian, even just a bit, fixes overconfidence in ReLU networks. In *International Conference on Machine Learning (ICML)*, 2020.

[7] Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms. *arXiv preprint arXiv:1708.07747*, 2017.

[8] Tarin Clanuwat, Mikel Bober-Irizar, Asanobu Kitamoto, Alex Lamb, Kazuaki Yamamoto, and David Ha. Deep learning for classical Japanese literature. *arXiv preprint arXiv:1812.01718*, 2018.

[9] Yarin Gal and Zoubin Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *International Conference on Machine Learning (ICML)*, 2016.

[10] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[11] Erik Daxberger, Agustinus Kristiadi, Alexander Immer, Runa Eschenhagen, Matthias Bauer, and Philipp Hennig. Laplace redux – effortless Bayesian deep learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[12] Beau Coker, Wessel P. Bruinsma, David R. Burt, Weiwei Pan, and Finale Doshi-Velez. Wide mean-field Bayesian neural networks ignore the data. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2022.

[13] Ali Unlu and Laurence Aitchison. Variational Laplace for Bayesian neural networks. *arXiv preprint arXiv:2011.10443*, 2020.

[14] Pavel Izmailov, Patrick Nicholson, Sanae Lotfi, and Andrew Gordon Wilson. Dangers of Bayesian model averaging under covariate shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[15] Kevin Raina and Tanya Schmah. Out-of-distribution detection from small training sets using Bayesian neural network classifiers. *arXiv preprint arXiv:2510.06025*, 2025.